extreme_picks

equity strategy | family: equity-baseline | status: validated

What this strategy does

Buy the top 10% of earnings surprises, short the bottom 10%, equal bets

Why it should work. Analysts and investors under-react to earnings news, so stocks with the most positive surprise keep drifting up and the most negative keep drifting down for weeks.

Design notes (from the strategy's source file).

Strategy 1 -- Extreme decile long/short.

The textbook PEAD trade: long the best earnings-surprise decile (D10), short the worst (D1), every event equal-weighted, each held for the number of days its (decile x size x book-to-market) cell's alpha takes to decay (per-cell holding windows, not a fixed horizon).

parameter	value
base_unit_fraction	0.003
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
beta_overlay	0.0

Performance

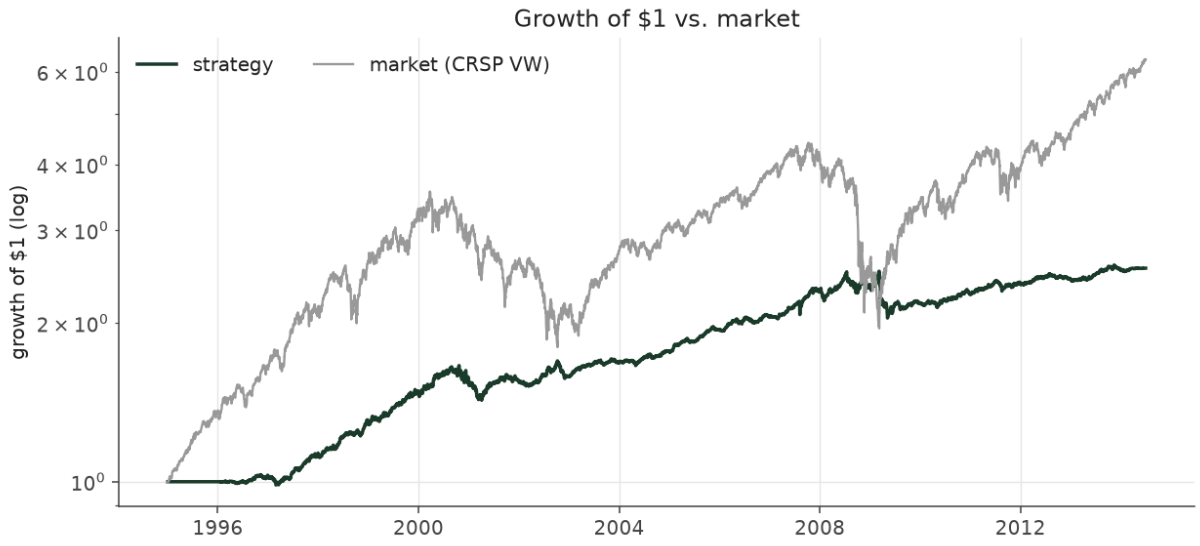
metric	extreme picks	market (CRSP VW)
Annualized return	4.91%	9.94%
Annualized volatility	5.32%	19.35%
Sharpe (NAV-pct)	0.93	0.59
Sortino	1.28	
Max drawdown	-18.5%	-55.5%
Calmar	0.27	
Final NAV ($10M start)	$25,432,376	
Worst day	-4.78%	
% positive days	52.0%	
Sharpe 1996-2001	1.28	
Sharpe 2002-2007	1.58	
Sharpe 2008-2013	0.28	

Correlation of daily returns with the market: **-0.00**.

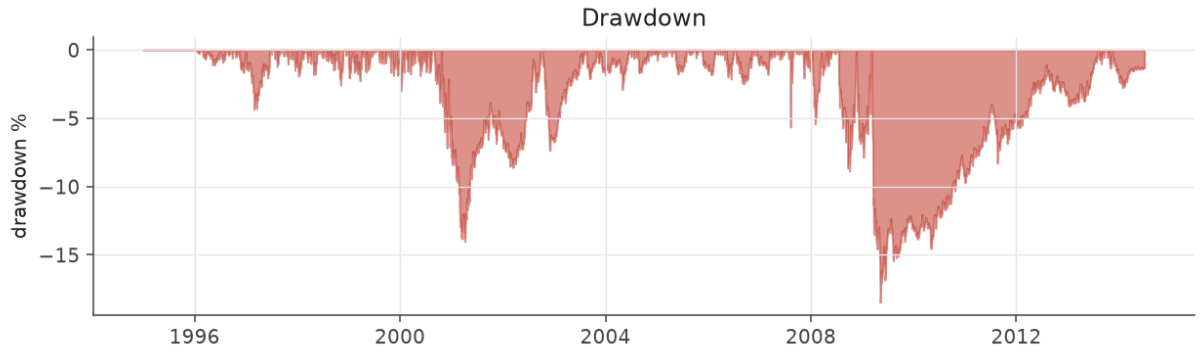
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.927

• best return among equity: `market_boost_max` (0.15); this one: 0.0491

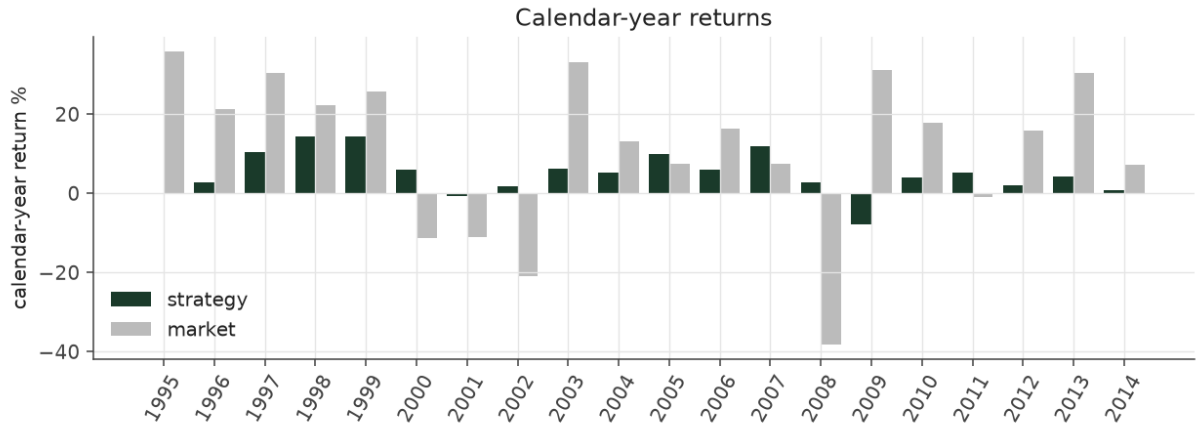
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.266



NAV (log scale)



Drawdown from running peak

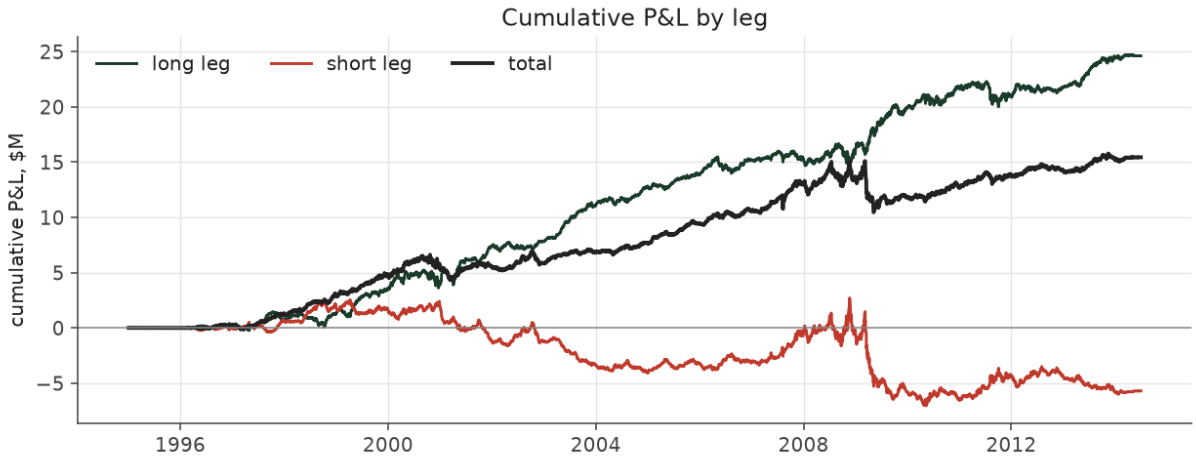


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	24.61	-5.70
P&L after costs ($M)	22.94	-7.50
of which common (average-event) component ($M)	13.75	-11.81
of which excess / PEAD component ($M)	10.86	6.11
excess minus costs ($M)	9.19	4.31
gross return per $ traded (bp)	275.07	-68.31
positions in the money (hit rate)	0.57	0.49
stand-alone Sharpe (leg only)	1.14	-0.20

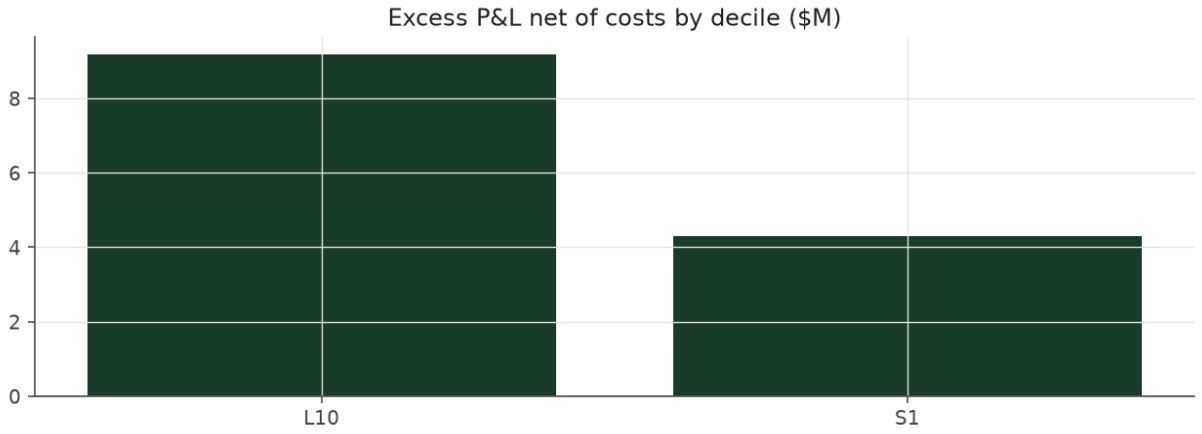
Daily P&L correlation between the two legs: **-0.68**.

• **The short leg loses money in absolute terms** (-5.7M before costs) while the long leg earns +24.6M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 13.8M / 11.8M. Stripping that common component, the surprise signal itself earned +10.9M on the long side and +6.1M on the short side.

• The two legs' daily P&L are -0.68 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.14 long / -0.20 short vs 0.93 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	17,174	894.62	22.94	10.86	1.67	9.19	68.1%	102.68
short	1	17,169	834.45	-7.50	6.11	1.80	4.31	31.9%	51.63



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,832	143.43	0.98	2.29	0.53	1.76	13.0%	122.42
2	7,159	359.29	4.21	5.42	0.99	4.43	32.8%	123.32
3	7,735	425.54	5.92	6.21	0.88	5.34	39.5%	125.39
4	7,731	424.71	2.45	2.08	0.66	1.42	10.5%	33.45
5	6,886	376.10	1.88	0.97	0.42	0.55	4.1%	14.66

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,357	95.01	2.64	3.33	0.47	2.86	21.2%	300.87
2	7,223	378.56	2.98	4.77	1.14	3.63	26.9%	95.94
3	7,834	436.75	6.04	6.66	0.87	5.79	42.9%	132.55
4	7,850	433.00	1.85	1.58	0.61	0.97	7.2%	22.45
5	7,077	385.66	1.93	0.65	0.39	0.26	1.9%	6.74

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	6,338	345.19	0.53	3.14	0.55	2.59	19.2%	75.09
41-60d	11,202	547.71	1.83	2.01	1.11	0.91	6.7%	16.54

61-90d	12,665	663.80	5.74	4.45	1.32	3.12	23.2%	47.06
>90d	4,138	172.37	7.34	7.37	0.50	6.87	50.9%	398.71

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	9,920	346.19	5.16	5.36	0.66	4.70	34.8%	135.76
2002-07	11,506	579.97	7.13	7.90	1.13	6.76	50.1%	116.60
2008-13	12,917	802.92	3.14	3.71	1.68	2.03	15.1%	25.30

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	6,718	324.80	3.30	2.21	0.64	1.57	11.6%	48.38
Chems	687	36.46	-0.23	-0.21	0.07	-0.28	-2.0%	-75.55
Durbl	869	42.30	0.12	0.33	0.09	0.24	1.8%	56.14
Enrgy	1,375	71.59	1.64	1.74	0.14	1.60	11.9%	223.38
Hlth	3,037	146.50	1.13	0.77	0.31	0.46	3.4%	31.05
Manuf	3,627	183.49	0.83	1.33	0.36	0.97	7.2%	53.10
Money	5,366	284.38	1.84	3.30	0.57	2.74	20.3%	96.26
NoDur	1,540	78.98	1.08	1.08	0.15	0.93	6.9%	117.52
Other	5,074	258.25	2.18	2.44	0.56	1.88	13.9%	72.78
Shops	3,547	176.30	2.46	2.68	0.35	2.33	17.2%	132.03
Telcm	1,164	56.60	0.77	0.93	0.11	0.82	6.1%	145.21
Utils	1,339	69.44	0.31	0.36	0.12	0.24	1.8%	34.24

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	11,517	585.34	2.46	2.98	1.06	1.92	14.2%	32.73
low vol	11,554	575.48	6.81	7.73	1.27	6.46	47.9%	112.21
mid vol	10,730	544.68	5.76	5.90	1.09	4.81	35.6%	88.25

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	4.91%	-0.00%
0.5%	1.0	4.70%	0.21%
1.0%	1.9	4.49%	0.42%
2.0%	3.8	4.04%	0.87%
5.0%	9.5	2.41%	2.50%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.